BRIEFING NOTE · SITUATIONAL AWARENESS

How disease-spread probabilities get their data

A weekly row of origin, destination, passenger volume, origin prevalence, and whether the destination was seeded in the next 14 days is the training set. COVID-19 is the only outbreak large enough to fill that table. SARS, H1N1, MERS, Ebola, and Zika check whether the same flight relationship still ranks who gets hit, after the chance an infectious person boards is changed.

Date 14 October 2026 **Prepared for** Kai **Scope** Data design for a world map of current outbreaks and importation risk

4,069	8.9M	0.97	0.34
Airports in the Brockmann passenger network	Passengers per day on that network	Simulated arrival-time R^2 from effective distance	Same simulation, geographic distance R^2

Source: Brockmann and Helbing, Science, 2013. The R^2 pair is their simulated pandemic. On real 2009 H1N1 and 2003 SARS country arrival dates, arrival time was still linear in effective distance. Residual variance in the simulation was more than 50 times smaller than with geographic distance.

Three numbers, three different inputs

Chance the pathogen is already in the selected country comes from local case reports and an under-reporting band. Chance an infected traveler arrives comes from prevalence times passengers. Chance it takes off locally comes from a branching process using that virus's reproduction number and cluster dispersion. Flight data enters only the second number.

The training row

A live flight API shows where aircraft are flying this week. The historical correlation needs passengers, case counts, and border rules on the same calendar. For every origin country, destination country, and week, the row contains four fields.

Field	What goes in it	Primary source
Origin prevalence	Reported cases divided by population, times an under-reporting factor. Testing changes who appears in the case file, so the factor has to travel with the row.	Our World in Data and the Johns Hopkins CSSE archive for COVID cases. WorldPop or UN population for the denominator.
Passengers that week	People moving from the origin to the destination, including the hub path when the direct route is thin.	US BTS and Eurostat where they observe the market. A gravity model elsewhere. OAG or IATA bookings if a paid feed is available.
Destination border rule	International-travel bans, quarantine, and screening in force that week.	Oxford COVID-19 Government Response Tracker, country-day, through 2022.
Label	1 if the destination recorded a first confirmed case, or a new viral lineage, in the next 14 days.	Official first-case dates to start. Nextstrain lineage introductions are the cleaner label, because the first official case mostly measures who was testing.

COVID-19 supplies on the order of 200 countries across many months of this panel. That is enough to estimate a handful of rates. Country-weeks are one branching process, repeated, and later waves are dominated by testing, vaccines, and closed borders, so the fit stays small and mechanistic. Disease identity enters as the chance of flying while infectious, the reproduction number, dispersion, and generation time. The passenger coefficient is what COVID estimates.

How the probability is calculated

Effective distance ranks who is next

If F(m→n) is passengers from m to n, the fraction of travelers leaving m who go directly to n is P = F(m→n) / F(m→anywhere). Edge length is 1 − ln(P). Distance between two countries is the shortest path, so a journey through Istanbul, Dubai, London, Singapore, or Atlanta counts.

Arrival time is that distance divided by one speed. Fit the speed on COVID first-wave arrival dates. Freeze it, and test whether H1N1 and SARS arrival order still follows the same ranking.

Importation, then establishment

Daily infected arrivals at a destination are the sum over origins of prevalence × passengers × α × (1 − screening). α is the chance an infectious person still boards, relative to anyone else. It is high for early COVID-19 and for Zika, and very low for Ebola.

Each arrival then starts a branching process with mean R and dispersion k for that virus. Most early COVID introductions died out because transmission was overdispersed. A busy route can make importation likely while a sustained local outbreak is still uncommon.

$$d(m, n) = 1 - \ln(P_{mn}) \quad \text{where} \quad P_{mn} = F_{mn} / F_m$$

D_{eff} = shortest path over those edge lengths

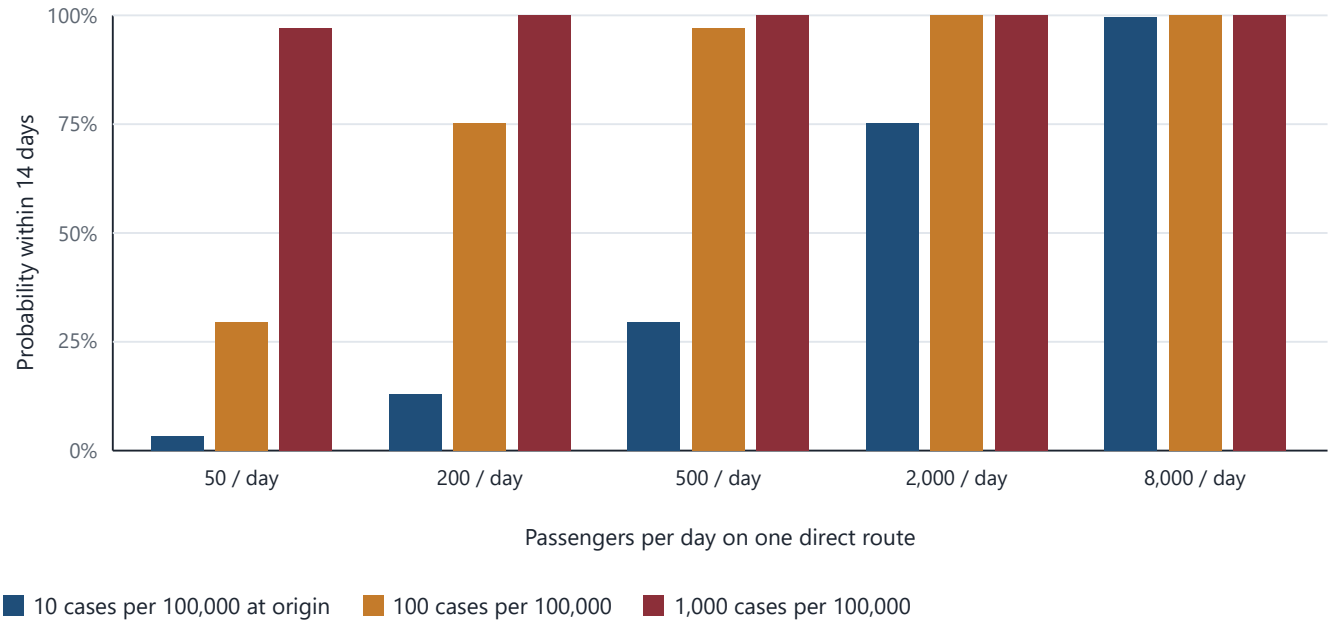
$$\text{expected arrivals / day} = \Sigma_{origins} (I/N) \times F \times \alpha \times (1 - \sigma)$$

$$P(\text{at least one arrival in T days}) = 1 - \exp(-\Sigma \text{ daily expectations})$$

σ is screening and border effectiveness, from 0 to about 0.9. OxCGRT supplies it in the historical fit. A live outbreak needs a manual policy flag until an equivalent tracker exists. “Already here” stays outside this formula: it is local incidence plus an under-reporting band.

Chance of at least one infected arrival in 14 days

One direct route. Infected people board at half the rate of everyone else. Screening is off. The bars are that scenario’s probability, not a fit to observations. Thin routes matter while prevalence is still low. Once prevalence is high, almost every scheduled route imports someone.



Illustrative Poisson model. Expected infected arrivals = prevalence × 0.5 × passengers × 14 days. Probability = 1 − exp(−expected arrivals). At 10 per 100,000, 50 passengers a day is about 3%, and 2,000 a day is about 75%. At 1,000 per 100,000, a 50-passenger route is already about 97%.

What to download

Outbreaks and the training labels

Dataset	Grain	Role	Access
WHO Disease Outbreak News API	Bulletin	Live disease, place, and narrative case counts	Free
ECDC threat reports, ProMED, HealthMap	Event	Earlier, noisier signals before a WHO bulletin	Free, limited
OWID covid-19-data and JHU CSSE	Country-day	COVID prevalence and first-case dates	Free
OxCGRT covid-policy-dataset	Country-day	Border closures and quarantine, the main confounder	Free, historical
Nextstrain, sequences via GISAID	Lineage introduction	Cleaner arrival label than the first official case	Viz open, sequences gated
FluNet, WHO SARS tables, Ebola sitreps, PAHO	Outbreak-week	Validation outbreaks with fewer country-weeks	Free, messy

WHO's Sitefinity API is at /api/news/diseaseoutbreaknews. Use TinyFish, or any extractor, only to pull case counts out of the prose when the API fields stay unstructured. TinyFish does not contain passenger history.

Passenger movement

Dataset	What it counts	Coverage	Access
OpenFlights	Airports and scheduled routes	Global topology, no passengers	Free
OpenSky COVID flights on Zenodo	Flights, from ADS-B	2019–2022. Weak coverage in parts of Africa and Asia	Free
US BTS T-100 International Market	Enplaned passengers	Monthly since 1990, only markets that touch the United States	Free
Eurostat aviation passenger tables	Passengers on airport pairs	Europe	Free
WorldPop Huang 2013 and Mao 2015 flows	Modeled passengers	Global, year 2010. Huang includes trips within two transfers	Free, stale
OAG or IATA booking data	True origin–destination passengers	Global, current, including connecting itineraries	Paid

Fit a gravity model on BTS and Eurostat: passengers from flight frequency, seats, city population, and distance. Use it on routes those agencies do not observe. For the live map, take this week's schedules from OpenSky or a schedule API, multiply by typical seats and a load factor, and replace that proxy when a booking feed exists.

GLEAM (gleamviz.org) is the academic system that already runs a metapopulation epidemic model on airline plus commuting flows. Copy the architecture. Its passenger matrix is not a bulk file an app can ship.

Inputs to the live formula

Input	Symbol	Where the number comes from
Origin prevalence	I / N	Reported cases ÷ WorldPop or UN population, times an under-reporting factor
Passenger flow	F	BTS or Eurostat where observed; gravity model elsewhere; live seats × load factor
Fly-while-infectious	α	Per-virus literature. High for early COVID-19 and Zika, very low for Ebola
Screening and borders	σ	OxCGRT for 2020–2022; a manual policy flag for a live outbreak
Reproduction number	R	Literature prior, replaced by local growth once cases exist
Dispersion	k	Cluster-size studies. COVID's k near 0.1 means most chains die
Generation time	—	Literature. Turns an arrival into a growth clock

How each virus is used

Outbreak	People flying while infectious	Job in the model	Records and the held-out check
COVID-19	High early on. A large share of transmission happened before people felt sick enough to stop flying.	Estimates how strongly passenger volume and border rules change importation. This is the only outbreak with daily cases, policy, and flight counts together.	OWID, JHU CSSE, OxCGRT, OpenSky 2019–2022, Nextstrain. Fit on the first wave. Test on Delta and Omicron introduction order.
H1N1 2009	High. People flew while infectious, and there was no global lockdown sitting on the signal.	Checks that effective distance, built from passenger flux, still ranks country arrival order.	WHO FluNet and published first-case dates. Rank correlation of arrival day versus effective distance. Refit nothing in the flight term.
SARS 2003	Medium. Infectiousness rose after symptoms, so fever screening removed more travelers than it did for early COVID-19.	Second arrival-order test, with a lower boarding probability.	WHO 2003 cumulative tables and published arrival dates. The same ranking should hold, with a lower α than COVID-19.
MERS	Low outside hospitals. Most amplification was nosocomial after a rare importation.	Checks that little community transmission does not turn airline volume into a pandemic probability.	WHO Disease Outbreak News archive, including South Korea in 2015. Importation can be real on a few routes. Establishment stays low almost everywhere else.
Ebola	Very low. People are usually too sick to fly during the highly infectious period, and exit screening was strict.	Negative control. Local R above 1 should still produce a low international establishment probability.	WHO situation reports, West Africa 2014–2016 and DRC 2018–2020. If the COVID-trained rate fires here, α is wrong.
Zika	High. Many infections were mild or asymptomatic, so infected travelers kept moving.	Separates seeding from establishment. Flights move people. Local mosquitoes decide whether it sticks.	PAHO reports and published introduction studies. Importation can be high in any well-connected city. Establishment tracks vector climate.

Worst-case last day to leave

Run the same formulas on an upper band: reported cases times a high under-reporting factor, the high end of R, screening set to zero, no new border closure, and this week's flight schedule left unchanged. The markers sit on the selected country.

Import marker

The first day the upper-band arrival probability crosses a threshold stated on the map. Fifty percent is a clear default. An infected traveler may land. A local epidemic is still a separate event.

Leave marker

The first day the upper-band establishment probability crosses that threshold, minus incubation time and the days required to arrive somewhere that is still below the threshold and still far in effective distance.

Stay marker

The day the selected country's own upper-band prevalence exceeds the places it is still possible to reach. Departing after that mostly moves exposure and can export cases.

> **What this date is**
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> It is a scenario threshold for the map. Official WHO and CDC travel notices stay the action layer. For an Ebola-like virus the leave marker should rarely be set by flights. The risk is being in the affected place already.

Build order

1. Pull WHO Disease Outbreak News from the public API. Parse leftover case counts from the narrative only when the fields are unstructured.
2. Download OpenFlights for the airport graph, BTS T-100 and Eurostat for observed passengers, and the WorldPop 2010 matrices as a global prior.
3. Fit passengers ≈ f(flights or seats, population, distance) on the observed markets. Score the fit on held-out BTS routes before using it anywhere else.
4. Attach OpenSky monthly flight counts for 2019–2022 so the COVID years carry the real volume shock, including the collapse and the partial return.
5. Join OWID prevalence and OxCGRT border rules. Regress first-wave arrival time on effective distance. Freeze the flight term.
6. Check rank order on H1N1 2009 and SARS 2003. Check that Ebola's international establishment probability stays low.
7. For the live map, scale this week's schedules by seats and a load factor. Show importation and establishment separately, with the three markers above.

Limits

- No free feed publishes current global origin–destination bookings. Until OAG or IATA is licensed, live passenger weights are an approximation anchored on BTS, Eurostat, and a gravity model.
- The open global matrices (Huang et al. 2013, Mao et al. 2015) describe 2010, and they are modeled flows.
- OpenSky counts flights, not people, and ADS-B reception is uneven.
- Official first-case dates are biased toward countries that tested early. Lineage introductions are the better label where they exist.
- OxCGRT is a historical file. A future outbreak needs its own border-policy coding, or the model will treat policy as if it were biology.
- The 14-day chart is a scenario with $\alpha = 0.5$ and no screening. It is there to show saturation. It is not an estimated risk for any current outbreak.

Sources

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- OpenFlights airport and route dumps. openflights.org/data
- GLEAM, Global Epidemic and Mobility model. gleamviz.org
- Nextstrain genomic builds. nextstrain.org. Underlying sequences are shared through GISAID.